KARTHICK

Python Developer | Backend & AI Systems

Chennai, India • +91 9952072343 • contact@karthicka.me • karthicka.me
github.com/akarthick-dev • linkedin.com/in/karthick-seven

PROFESSIONAL SUMMARY

Python Developer focused on backend development, building RESTful APIs with FastAPI, and implementing Generative AI pipelines. Experienced in developing Retrieval-Augmented Generation (RAG) systems using LangChain and ChromaDB, creating autonomous terminal agents using FastMCP (Model Context Protocol), and designing automated workflows with n8n. Skilled in core Python development, MySQL database integration, and building reliable developer tools.

EDUCATION

Bsc Computer Science — AM jain college 2023 – 2026 | Chennai, India

TECHNICAL SKILLS

Backend & APIs:	FastAPI, RESTful APIs, Pydantic
Languages & DB:	Python (OOP), MySQL, ChromaDB (Vector DB)
AI & Automation:	FastMCP (Model Context Protocol), LangChain, RAG Pipelines, Vector Search, Ollama, n8n
Audio & Speech:	OpenAI Whisper / faster-whisper (STT), ElevenLabs API (TTS), Groq API
Developer Tools:	Git, GitHub, Linux
Methodologies:	Agile, Waterfall Model

INTERNSHIP

Gen AI Intern — CFRS Chennai, India

- Researched and built Generative AI pipelines and prompt engineering workflows for automated data processing.
- Implemented Retrieval-Augmented Generation (RAG) components using LangChain and vector databases for contextual query resolution.
- Engineered backend API endpoints using FastAPI to serve model outputs and integrate LLM inference into downstream services.

TECHNICAL PROJECTS

Terminal AI Agent with MCP | Python, FastMCP, CLI | GitHub Repository Open Source

- Engineered an autonomous terminal AI assistant in Python powered by FastMCP (Model Context Protocol) for structured tool calling.
- Implemented custom MCP tools for file system management (read_file, write_file, list_files), regex-based text search, and shell execution.
- Integrated strict security guardrails with interactive user approval flags and execution timeouts to prevent unsafe terminal operations.

Retrieval-Augmented Generation (RAG) | Python, FastAPI, LangChain, ChromaDB | GitHub: RAG Open Source

- Architected an end-to-end document question-answering microservice using FastAPI, LangChain, and ChromaDB for semantic retrieval.
- Implemented text chunking with RecursiveCharacterTextSplitter and dense vector embeddings, achieving sub-100ms similarity lookups.
- Designed typed RESTful endpoints (/upload, /ask) with Pydantic request validation and citation-backed generation to eliminate hallucinations.

Real-Time 3D AI Companion | Python, FastAPI, Three.js, Groq, ElevenLabs | sana.karthicka.me Live Project

- Built an interactive 3D AI companion from scratch inspired by Grok Ani, uniting real-time voice, WebGL avatar rendering, and LLM reasoning.
- Engineered a sub-second voice pipeline orchestrating faster-whisper, Groq inference, and ElevenLabs streaming synthesis.
- Implemented dynamic emotion tagging driving live 3D avatar expressions and gestures with persistent session memory across multi-turn dialogues.

ACHIEVEMENTS & CERTIFICATIONS

1st Place — AI Agent Automation Challenge | Inter-College Technical Symposium Winner

- Secured 1st place among 15+ competing teams by designing and demonstrating an automated agentic workflow using n8n and Python integrations.

HACKFORGE 2025 — National Level 24-Hour Hackathon | DG Vaishnav College & Tech Knots Dec 2025

- Selected and participated in a 24-hour national hackathon, collaborating on rapid software prototyping and solution delivery.

Cyber Security Certification (Grade A) | IBM SkillsBuild & ICT Academy Mar 2026

- Completed certification course on cybersecurity fundamentals, network security protocols, and system defenses.